

SCENARIO

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NAME	MATRIC NO	GROUP
IRDINA NADHIRAH BINTI BAHANUDDIN	22C18G026	2

PROBLEM:

Write a program that ask user to input the height in metres and weight in kilogram, then the program will calculate the Body Mass Index (BMI). The formula of BMI is as below:

$$\text{BMI} = \text{weight} / (\text{height} * \text{height})$$

Afterwards, the program will display the category of the person's body based on the following table:

SAMPLE OUTPUT:

Sample Output:

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IPO CHART		
INPUT	PROCESS	OUTPUT
Weight	BMI=Weight/(Height*Height)	Print_BMI
Height	if (BMI<18.5) Print_Category_Underweight; else if (BMI>=18.5 && BMI<=25) Print_Category_Normal; else if (BMI>=25.1 && BMI<=30) Print_Category_Overweight; else if (BMI>30); Print_Category_Obesity; else Print_Invalid_Input;	Print_Category

PSEUDOCODE:

Start

Prompt Weight

Read Weight

Prompt Height

Read Height

$BMI = \text{Weight} / (\text{Height} * \text{Height})$

Print_BMI

if (BMI < 18.5)

 Print_Category_Underweight;

else if (BMI >= 18.5 && BMI <= 25)

 Print_Category_Normal;

else if (BMI >= 25.1 && BMI <= 30)

 Print_Category_Overweight;

else if (BMI > 30);

 Print_Category_Obesity;

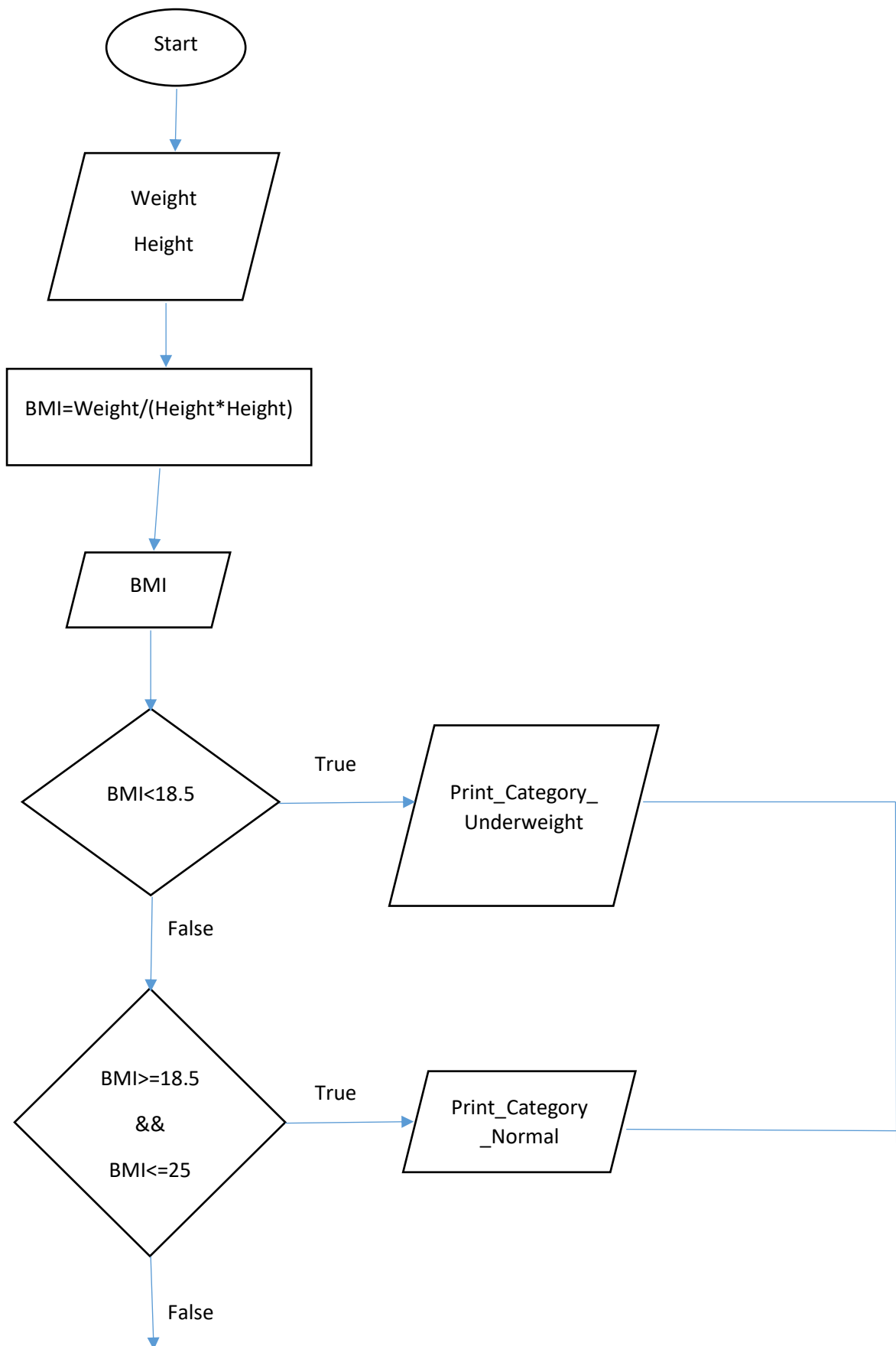
else

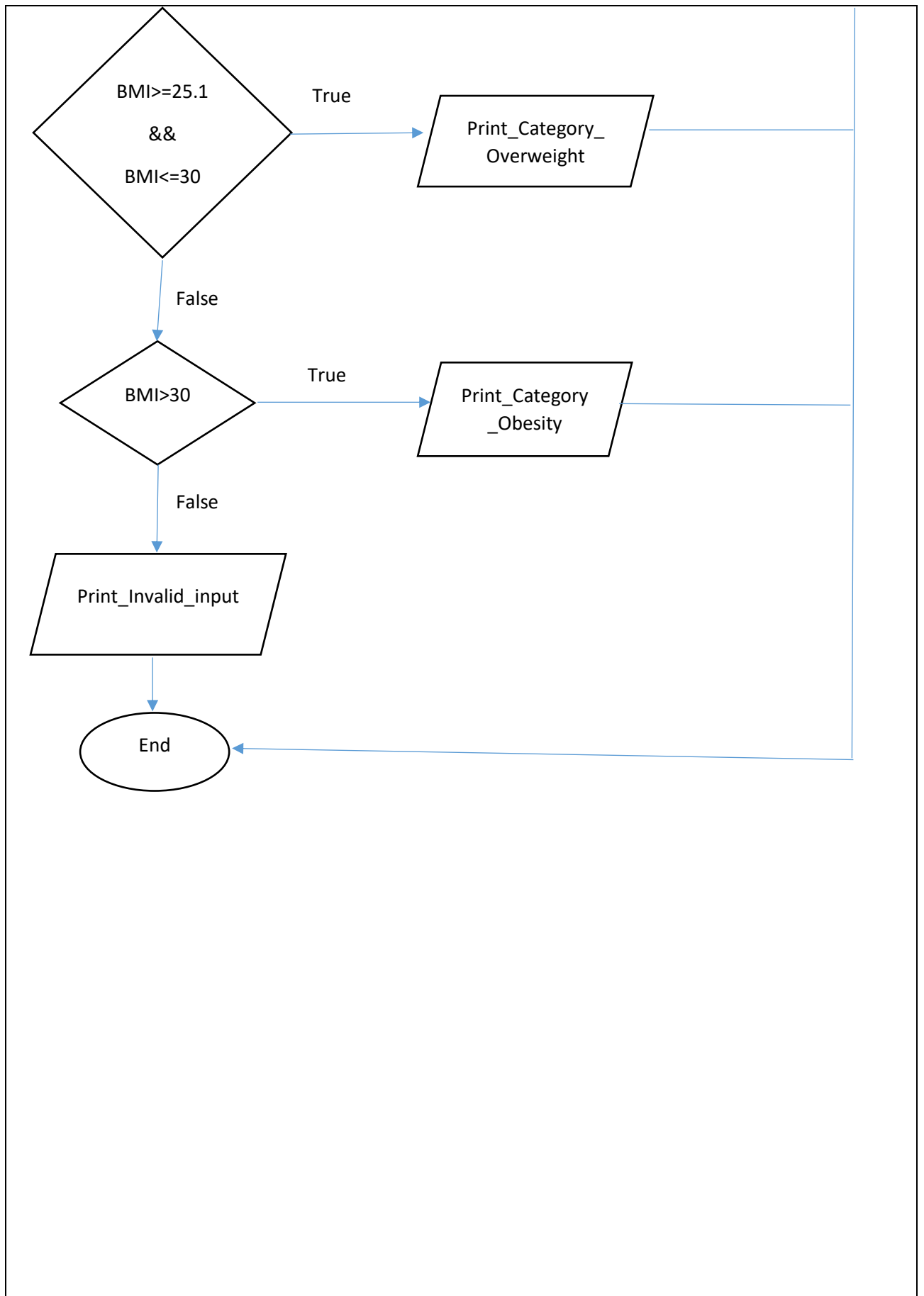
 Print_Invalid_Input;

Print_Category

End

FLOWCHART:





C++ PROGRAM:

```
#include <iostream>

using namespace std;

int main()
{
    float Weight;
    float Height;
    float BMI;

    cout<<"Please enter your weight in kilogram = ";
    cin>>Weight;

    cout<<"\nPlease enter your height in metres = ";
    cin>>Height;

    BMI=Weight/(Height*Height);

    cout<<"\nYour BMI is "<<BMI;

    if (BMI<18.5 )
        cout<<"\nCategory Underweight";
    else if (BMI>=18.5 && BMI<=25)
        cout<<"\nCategory Normal";
    else if (BMI>=25.1 && BMI<=30)
        cout<<"\nCategory Overweight";
    else if (BMI>30)
        cout<<"\nCategory Obesity";
    else
        cout<<"\nInvalid input!";

    return 0;
}
```